

Paid to Hold the Bag

Liquidity provision is a high-VIX premium — at the daily horizon and at the opening auction — and that is exactly why it is hard to size

Gator Quant Hacks 2026 · Track 03 Systematic Trading · October 2026 · All numbers: `python run_all.py --oos` in the public repo · Hypotheses pre-registered in HYPOTHESIS.md (commit 0d0093d, 2026-10-03 10:54 UTC) and HYPOTHESIS_B.md (commit fbd355a, 11:53 UTC), each before its first backtest

1. Summary

Short-term reversal is the return to supplying liquidity. Before looking at any result we hypothesised that this premium is larger when liquidity-provider capital is scarce, proxied by yesterday's VIX, and tested the one mechanism on two independent instruments. **Module A** (36 years of Kenneth French daily reversal portfolios): the expected reversal return rises with lagged VIX (Newey–West $t = 2.4$), and since 2004 the large-cap premium exists *only* in the top VIX quintile, 14 bps/day when $VIX > 24$, about zero otherwise; on the 44 held-out days above that level it earned 52 bps/day, on the other 455 nothing. The pre-registered rule that *levers* into high VIX fails: the book's volatility rises 3.6× across the same quintiles, so more dollars buy the same Sharpe. **Module B** takes the lesson to the opening auction. Fading overnight gaps in S&P 500 names from open to close earns 18 bps/day gross ($t = 12$, gross Sharpe 3.3), 35 bps/day in the top VIX quintile versus 12 elsewhere. At 5 bps per dollar traded it is a market maker's premium: always-on nets nothing, and the pre-registered rule — *participate* only when $VIX_{t-1} \geq 20$, rather than lever — turns a -0.26 Sharpe into $+0.12$ in-sample and -1.83 into $+0.24$ on the held-out two years, when the top-VIX state paid 56 bps/day and every other state zero. The premium has decayed: 33 bps/day before 2013, 8 after. What survives for a non-market-maker is a constant-notional large-cap reversal book (net Sharpe ≈ 0.4 , break-even 14 bps, \$100M–\$500M capacity) and, for a desk with 2.5 bps auction execution, a VIX-gated gap fade with Sharpe ≈ 1 . All net of costs; all 62 variants reported.

2. Economic hypothesis (written before results)

Who is on the other side. Someone who must trade *now*: a fund meeting redemptions, a forced deleverager, an investor reacting to overnight news with a market-on-open order. They pay, through a temporarily displaced price, for a counterparty; the price then reverts. A reversal book, daily or intraday, is a systematic way of being that counterparty.

Why it should depend on VIX. Liquidity providers hold inventory risk. When volatility is high their VaR limits bind, margins rise and capital is withdrawn, so the price of immediacy rises (Grossman–Miller 1988; Brunnermeier–Pedersen 2009; Nagel 2012). The edge persists because the capital that would compete it away is what is scarce when the premium is highest; at the open the same logic applies to the overnight gap (Berkman et al. 2012; Lou, Polk & Skouras 2019).

Pre-registered predictions. Module A — P1: slope of the daily reversal return on VIX_{t-1} positive, NW $t > 2$; P2: mean rises across lagged-VIX quintiles; P3: $w_t = \min(VIX_{t-1}/20, 3)$ beats $w = 1$ net; P4: alpha survives FF5 + momentum; P5: neighbouring parameters agree. Module B, written after A's result and before any stock data was downloaded — B1: gross gap-fade spread positive, $t > 2$; B2: slope on VIX_{t-1} positive; B3: at 5 bps, a $VIX \geq 20$ participation rule beats always-on, with positive net mean; B4: both legs earn it (long leg only would suggest survivorship); B5: plateau over thresholds {15, 20, 25} and {quintile, decile}. P3 and half of B3 failed; kill criteria were written for each.

3. Data and universe

All public, no keys. **A:** Kenneth French Data Library (CRSP-based, point-in-time, through Aug 2026): the daily **ST_Rev** factor (long the bottom 30% on day -20 to -1 return, short the top 30%, **re-formed daily**);

the daily 6 size $\times$ prior-return portfolios, from which we build the **big-cap reversal leg** BIG Lo – BIG Hi; FF5 + momentum. **B:** daily adjusted open and close for the current S&P 500 constituents (Wikipedia list snapshotted and committed; prices from Yahoo Finance via yfinance, cached, not committed). Each stock enters **only from its index-addition date** (326 eligible names per day in-sample, 480 recently), so none is traded before it was a large cap; removed names are absent, so the universe has survivorship bias — disclosed, and tested by B4. **Both:** FRED CBOE VIX close, aligned by a strict as-of merge so the position on day t sees only VIX closes dated before t (unit-tested).

Samples: A 1990-01-02 $\rightarrow$ 2026-08-31 (8,732 in-sample days); B 2005-01-03 $\rightarrow$ 2026-08-31 (4,949). **Held out for both: 2024-09-01 $\rightarrow$ 2026-08-31** (the 2-year cap binds; 500 days), evaluated once per module, each in its own commit after the in-sample commit.

4. Methodology

Module A book and rule. One unit is \$1 long losers / \$1 short winners. $w_t = \min(\text{VIX}_{t-1}/20, 3)$: 20 is the textbook long-run VIX, so $w = 1$ at VIX 20 and the rule is comparable to constant exposure; 3 is a leverage limit; nothing is fitted. **Costs:** a 20-day formation window implies 12–15% of each side turns over daily; we assume $\tau = 14\%$ one-way, dollars traded $= 2w_t\tau + 2|\Delta w_t|$, and cost per dollar $c_t = c_{\text{base}} \times \text{VIX}_{t-1}/20$ (spreads widen with VIX; a flat-cost model flattens a VIX-timed rule), $c_{\text{base}} = 5$ bps large-cap, 10 bps all-cap; flat, $\times 2$ and break-even costs are also reported.

Module B book and rule. For each stock, $\text{gap} = \text{Open}_t / \text{Close}_{t-1} - 1$ minus the cross-sectional median gap; sort into quintiles at the open; \$1 long the most-gapped-down quintile, \$1 short the most-gapped-up, equal-weighted, **entered at the official open and closed at the official close** — never an overnight position. Rule: on iff $\text{VIX}_{t-1} \geq 20$, benchmark always-on. Four dollars are traded per day per \$1-per-side, so the toll is $4 \times 5 \text{ bps} \times \text{VIX}_{t-1}/20 \approx 20 \text{ bps/day}$; $c_{\text{base}} \in \{2.5, 10\}$ and the break-even are reported. Stock-days with $|\text{gap}| > 20\%$ (spin-off artefacts) are excluded using open information only; without the filter results are identical to 0.1 bps.

Tests and budget. Slopes by OLS with Newey–West (10 lags); rule comparisons by a spanning regression of rule on benchmark (HAC t on the intercept); A's alpha by FF5 + Mom regression. Pre-registered grids: A, normaliser $\{15, 20, 25\} \times \text{cap} \{2, 3, 4\} \times \text{universe} \times \{\text{linear, regime}\} = 36$, plus 2 post-hoc rules (§5.4) = 38 trials entering the deflated Sharpe (Bailey & López de Prado 2014); B, threshold $\{\text{off}, 15, 20, 25\} \times \{\text{quintile, decile}\} \times c_{\text{base}} \{2.5, 5, 10\} = 24$. **62 variants, all reported.**

5. Results

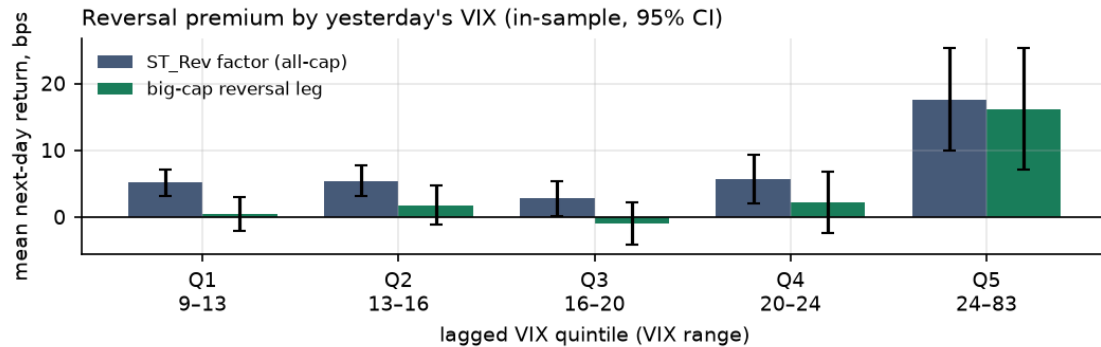
5.1 Module A: the premium depends on yesterday's VIX (P1 passes, P2 passes in shape; both replicate)

Mean next-day return, bps	slope / VIX pt	NW t	Q1	Q2	Q3	Q4	Q5 (VIX > 24)
Big-cap leg, in-sample 1990–2024	0.89	2.42	0.4	1.8	−1.0	2.2	16.1 $\pm$ 9.1
ST_Rev, in-sample	0.78	2.45	5.2	5.4	2.8	5.7	17.6 $\pm$ 7.6
Big-cap leg, out-of-sample	2.94	2.50	–	6.9	−4.3	−13.1	52.4 $\pm$ 50.7
ST_Rev, out-of-sample	2.83	3.10	–	3.1	−1.9	−8.2	47.3 $\pm$ 44.3
Big-cap leg, 2004–2024 only	0.82	1.66	−3.3	1.2	−4.5	0.2	14.0

$\pm$ is a 95% CI. 1,746 days per in-sample quintile; edges 13.2 / 16.1 / 19.6 / 24.3. Out-of-sample binned on those edges (44 days in Q5; Q1 has one day, omitted). Constant-book net Sharpe 1.02 (big-cap) / 2.25 (ST_Rev) in 1990–2003, −0.02 for both in 2004–2024.

P1 passes and replicates with the same sign and a larger slope. P2 is not monotone: flat across quintiles 1–4, then a jump. For large caps the premium is *only* in quintile 5, and the sub-period row shows why: after decimalisation and electronic market making the unconditional premium went to zero, as the hypothesis

says it should when liquidity-provider capital becomes abundant; what survives is the part earned when that capital withdraws.



5.2 Module A's rule: more dollars, not more Sharpe (P3 fails, P4 passes)

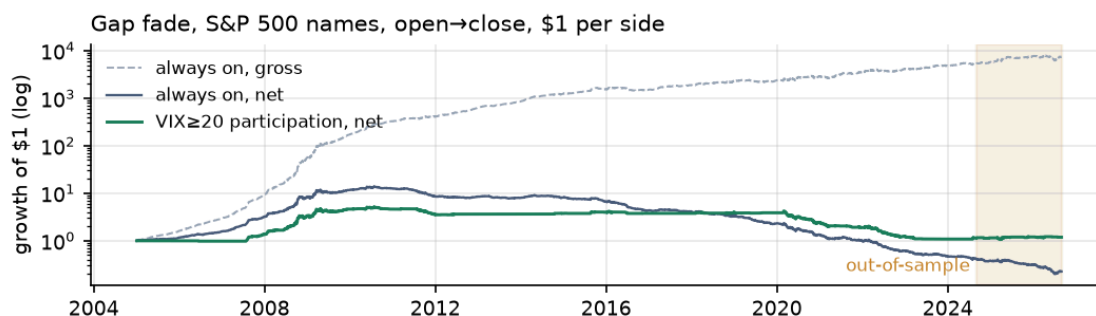
Net of costs	Constant, IS	VIX-timed, IS	Timed, flat costs	Timed, costs ×2	Constant, OOS	Timed, OOS
Big-cap: ann. return / vol	6.4% / 17.2%	10.9% / 30.3%	11.5% / 5.6% / 30.3%	5.6% / 30.3%	2.7% / 20.5%	8.0% / 23.8%
Big-cap: net Sharpe (gross)	0.37 (0.57)	0.36 (0.53)	0.38	0.18	0.13 (0.29)	0.34 (0.54)
Big-cap: max DD / worst month / skew	-48% / -17% / +1.2	-77% / -34% / +4.1			-27% / -8%	-24% / -9%
ST_Rev: net Sharpe (gross)	0.82 (1.30)	0.51 (0.93)	0.56	0.10	0.04 (0.45)	0.21 (0.70)
Break-even c_{base} , big-cap / ST_Rev	14.3 / 26.9 bps	15.3 / 22.4 bps				

The timed rule lifts big-cap gross return from 9.9% to 16.2% at the same mean exposure, as predicted, but volatility goes from 17% to 30%: the book's own volatility scales with VIX, so sizing linearly in VIX scales risk quadratically. The spanning alpha of timed on constant is +0.24 bps/day ($t = 0.3$) for big caps and **-2.35 bps/day ($t = -3.0$)** for ST_Rev, whose calm-state premium the rule discards. P3 fails. P4 passes: FF5 + Mom alpha 5.8%/yr ($t = 2.1$) for the constant big-cap book, 11.1%/yr ($t = 4.8$) for ST_Rev; market beta 0.2, $R^2 \approx 0.1$. The grid is a plateau (0.34–0.39; P5 passes) and luck alone would give a best-of-38 Sharpe of 0.24. Out of sample the timed rule beat constant, but the spanning alpha has $t = 1.1$ after failing on 35 years; we do not count it as support for P3: the premium arrived on the 44 high-VIX days and a rule long VIX happened to be large on them (equity curves in the appendix).

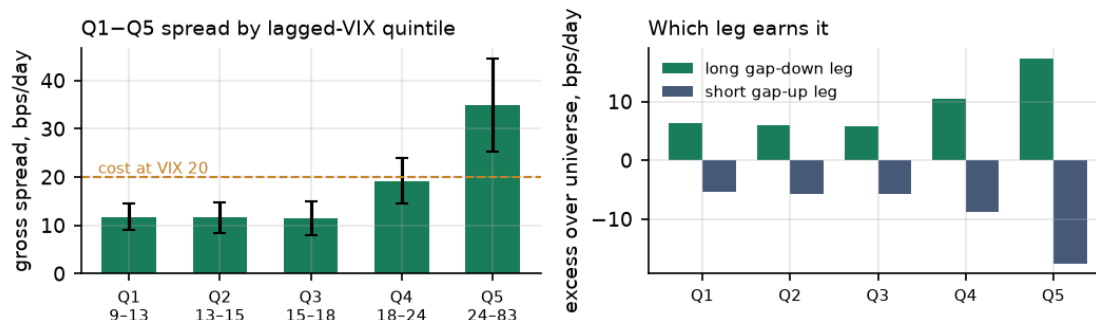
5.3 Module B: the opening auction pays the same premium, in the same states (B1, B2, B4, B5 pass; B3 half)

Q1 – Q5 gap-fade spread, open → close	gross bps/day (NW t)	slope per VIX pt (t)	VIX Q1 ... Q5, bps	net Sharpe @5 bps: always-on / gated	spanning t	break-even c_{base}	@2.5 bps: always / gated	days on
In-sample 2005–2024	17.8 (12.1)	1.39 (3.8)	12 · 12 · 11 · 19 · 35 ± 10	-0.26 / +0.12	2.6	4.6 / 5.3 bps	1.51 / 1.12	33%
2005–2012	32.6 (12.3)	1.69 (3.7)	24 → 59	1.84 / 1.19		7.6 / 7.6		
2013–2024	7.6 (5.6)	0.26 (0.8)	5 → 11	-2.06 / -1.00		2.2 / 2.3		
Out-of-sample 2024-09 → 2026-08	6.7 (1.4)	3.98 (3.3)	- · 0 · 0 · 4 · 56 ± 43	-1.83 / +0.24	2.0	1.8 / 5.9	-0.40 / 0.92	24%

Equal-weighted quintiles on the gap relative to the median gap; \$1 per side; cost $4 \times c_{\text{base}} \times \text{VIX}_{t-1}/20$ per day. OOS VIX bins use in-sample edges (Q1 empty). Legs, excess over the universe's open $\rightarrow$ close return: long gap-down +9.2 bps ($t = 11.8$), short gap-up -8.6 bps ($t = -11.2$) in-sample; +4.4 / -2.3 out of sample. Decile sort: net Sharpe at 5 bps 0.97 always-on / 0.89 gated in-sample, -1.23 / 0.53 out of sample.



Gross, this is a 3-Sharpe machine, and it is the mechanism of \$5.1 at a shorter horizon: 35 bps/day in the top VIX quintile against 12 in the rest, both legs symmetric (B4 passes; survivorship is not the story), slope $t = 3.8$. Net, at 5 bps per dollar traded, the ≈ 20 bps/day toll eats the unconditional premium (always-on nets -0.26), and the participation rule does what it was built to do — pays the toll only when the premium exceeds it, lifting Sharpe to +0.12 with spanning $t = 2.6$ — but its net mean is not significantly positive ($t = 0.5$): **at 5 bps this is a market maker's premium**. Two facts matter more than the headline. The premium is decaying: 33 bps/day and a 7.6 bps break-even before 2013, 8 bps and 2.2 bps since, as electronic market making in the auctions caught up; a 2.5 bps executor earned Sharpe 1.5 over the full sample and roughly zero since 2013. And the held-out two years are the sharpest state-dependence we have seen anywhere: unconditional gross fell to 6.7 bps, the VIX slope rose to 4 bps per point ($t = 3.3$), the 50 days with $\text{VIX}_{t-1} > 24$ (the spring 2025 and spring 2026 spikes) paid **56 bps/day** and every other state paid zero. Always-on lost 30% net; the gated book made 2.5% with a 9% drawdown, Sharpe 0.9 at 2.5 bps.



5.4 Post-hoc, disclosed: sizing Module A by risk instead (2 extra trials)

The natural fix for §5.2 is to size by expected return per unit of variance: $w_t \propto \max(E[r_t | \text{VIX}_{t-1}], 0) / \sigma_t^2$, with $E[r | \text{VIX}]$ from an expanding regression on days before t (5-year burn-in) and σ the trailing 63-day vol, against a vol-targeted book (Moreira–Muir 2017) and constant exposure on the same days (big-cap leg, 1995–2024, net Sharpe / max DD / worst month): constant **0.31** / -48% / -17%; vol-targeted **0.01** / -70% / -14%; mean-variance **0.19** / -74% / -53%. Vol targeting, the default risk tool, **destroys the strategy**: it de-risks exactly when the premium is paid. The mean-variance rule's -53% month is March 2020: the forecast said "premium high", the rule went to the 3× cap, and reversal kept losing as the crash continued (out of sample: 0.24× average exposure, Sharpe 0.45, two years). This is why Module B pre-registered participation rather than leverage: the premium is large in high-VIX states *because* the

outcomes there are wild, and with ~1,750 such days in a few dozen episodes no rule can estimate the conditional moments well enough to lever safely.

6. Risk management

What the books are. A: long/short equity, residual market beta 0.2, skew +1.2 and excess kurtosis 23 at constant notional; the timed version has skew +4 and kurtosis 127, a P&L made of a few enormous days. B: dollar-neutral, equal-weighted, flat at every close, so no overnight, gap or financing risk; its risk is one session's cross-sectional dispersion (worst day -6.6%, 9 Nov 2020, the vaccine rotation: those gaps were information). **Limits:** gross $\leq 2\times$ capital (3 $\times$ was A's pre-registered cap); per-name $\leq 2\%$ of gross with a mega-cap concentration cap; sector-neutral legs; a futures hedge taking beta to zero. **De-risking rule, set in advance: none that depends on realised vol**, since §5.4 shows that is the one thing that kills the premium. *Participation* in the state is the risk tool instead: B is off two-thirds of the time and, when on, sized so the pre-registered worst month at 1 $\times$ (A -17%; B -16%) and drawdown (A -48%; B gated -79%, a slow bleed from 2010 to 2023 through 2011 and 2020–22, when it was on and the premium had thinned) are survivable — a 5%-per-month risk budget implies 0.3 $\times$ capital for either. **Tail and regime:** the book loses when the hurried trader knew something (2009, 2020). Those are the risks being insured. In quintiles 1–4 the modern premium is zero, so a non-market-maker is paid nothing for the risk it runs there.

7. Liquidity and capacity

A turns over 70 $\times$ its capital a year. One-way cost = half-spread + square-root impact, $c(K) = 1.5 \text{ bps} + \sigma_{\text{name}} \sqrt{\text{participation}}$, with ≈ 500 large-cap names, value-weighted ADV $\approx \$200\text{M}$, single-name daily vol 2% at VIX 20 (src/gqh/capacity.py): \$10M $\rightarrow$ 2.6 bps and Sharpe 0.47; \$100M $\rightarrow$ 4.8 bps, 0.38; \$500M $\rightarrow$ 9.0 bps, 0.21; \$1.5B $\rightarrow$ 14.3 bps, 0.00. Our 5 bps assumption is about \$100M; the edge is half gone near \$500M. B trades only in the two auctions, at one price with no spread to cross, and its constraint is the open, which clears only ~1–2% of a name's daily volume (the close clears ~10%). With ~80 names per leg, \$25M of book is ~\$150k per name, about 5% of a \$3M opening auction in a \$200M-ADV name; \$100M would be ~20%, where impact sets the cost. B is a desk-scale trade — tens of millions, not hundreds; 5 bps is our headline and 2.5 bps an execution desk's number. A market maker who *earns* the half-spread faces a negative first term in both books and the gross Sharpes (0.57 and 3.3) are on offer, which is why both premia belong to market makers.

8. Limitations and next steps

(i) The French portfolios are a test instrument, not a product; B's universe is survivorship-biased (removed names are missing), though the symmetric legs say the result is not drift. (ii) A's turnover and both cost models are assumed, not observed, and stressed; a desk's fill data would replace them. (iii) B assumes execution at the official open from orders placed on pre-market prints; that slippage sits in the cost assumption, and part of the premium may live inside the first minutes. (iv) VIX is one proxy for intermediary capital, and the top-quintile effect rests on ~1,750 in-sample and 44–50 held-out days: the sign replicated with 2–4 $\times$ the slope, but the 52–56 bps carry ± 45 bps intervals. (v) "VIX > 24" was deliberately *not* turned into a rule — that would be the snooping the brief warns about — and Yahoo data can be revised, so the download is cached and the constituent file frozen. **Next:** non-volatility proxies for intermediary capital (dealer leverage, funding spreads) so sizing can be separated from vol; signed auction imbalance; a Q5-only book on a sealed window before believing 50 bps.

References (outside the page limit)

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Appendix (optional, outside the page limit)

A1. Module A, all 36 pre-registered variants, net Sharpe in-sample. Big-cap, linear rule, rows cap {2,3,4} × cols normaliser {15,20,25}: 0.39 0.37 0.36 / 0.37 0.36 0.35 / 0.36 0.35 0.34; regime rule 0.39. ST_Rev: 0.62 0.57 0.54 / 0.55 0.51 0.49 / 0.51 0.48 0.48; regime rule 0.47. results/tables/sensitivity_grid.csv.

A2. Module B, all 24 pre-registered variants, net Sharpe in-sample (rows c_{base} 2.5 / 5 / 10 bps; cols always-on, VIX ≥ 15, 20, 25). Quintile: 1.51 1.19 1.12 0.87 / -0.26 -0.20 0.12 0.17 / -3.80 -2.97 -1.88 -1.22. Decile: 2.32 1.88 1.65 1.27 / 0.97 0.81 0.89 0.75 / -1.77 -1.35 -0.67 -0.33. Out of sample, quintile: -0.40 -0.29 0.92 1.04 / -1.83 -1.64 0.24 0.68 / -4.73 -4.33 -1.13 -0.07. results/tables/gapfade_sensitivity_grid*.csv, results/REPORT_B.md.

A3. Big-cap leg, net return by year, timed / constant (%). 1990–99: 6/3, 13/14, 9/13, 3/4, 15/24, 15/25, 18/21, 11/12, 34/17, -5/-3. 2000–09: 34/23, 69/50, 56/28, 16/9, 0/0, -8/-13, -3/-5, -5/-9, 61/31, -19/0. 2010–24: 4/2, 1/-2, -11/-11, 1/2, 2/0, -2/-7, 7/12, -3/-5, -2/-4, 11/13, -22/-5, -7/-12, 0/-3, -15/-12, 2/-4. **Gap fade, net by year, gated / always-on (%):** 2005–09: 0/17, -2/35, 41/106, 142/159, 39/41; 2010–14: 0/7, -23/-32, 1/-1, 0/-6, 3/2; 2015–19: 6/-18, -5/-35, 0/-5, 4/-20, -1/-30; 2020–24: -37/-42, -22/-24, -35/-40, -12/-23, 5/-12.

A4. Honesty log. Module A: hypothesis commit 10:54 UTC; in-sample commit 11:11 with OOS locked; OOS evaluated once and committed 11:12. Module B: hypothesis and frozen constituent list committed 11:53:57, before any price download; in-sample committed 12:01:40; OOS evaluated once, committed 12:02:08. The sub-period split of B (2005–2012 / 2013–2024) was decided after seeing the by-year table and is labelled descriptive. One bug fixed after A's first in-sample run: factor betas were in mixed units; alphas and t-stats unaffected. 62 variants in total. The practitioner material we were given (an intraday gap-fade strategy matrix and ML-ops code) contained no price or trade data and was used only to motivate Module B's design.

